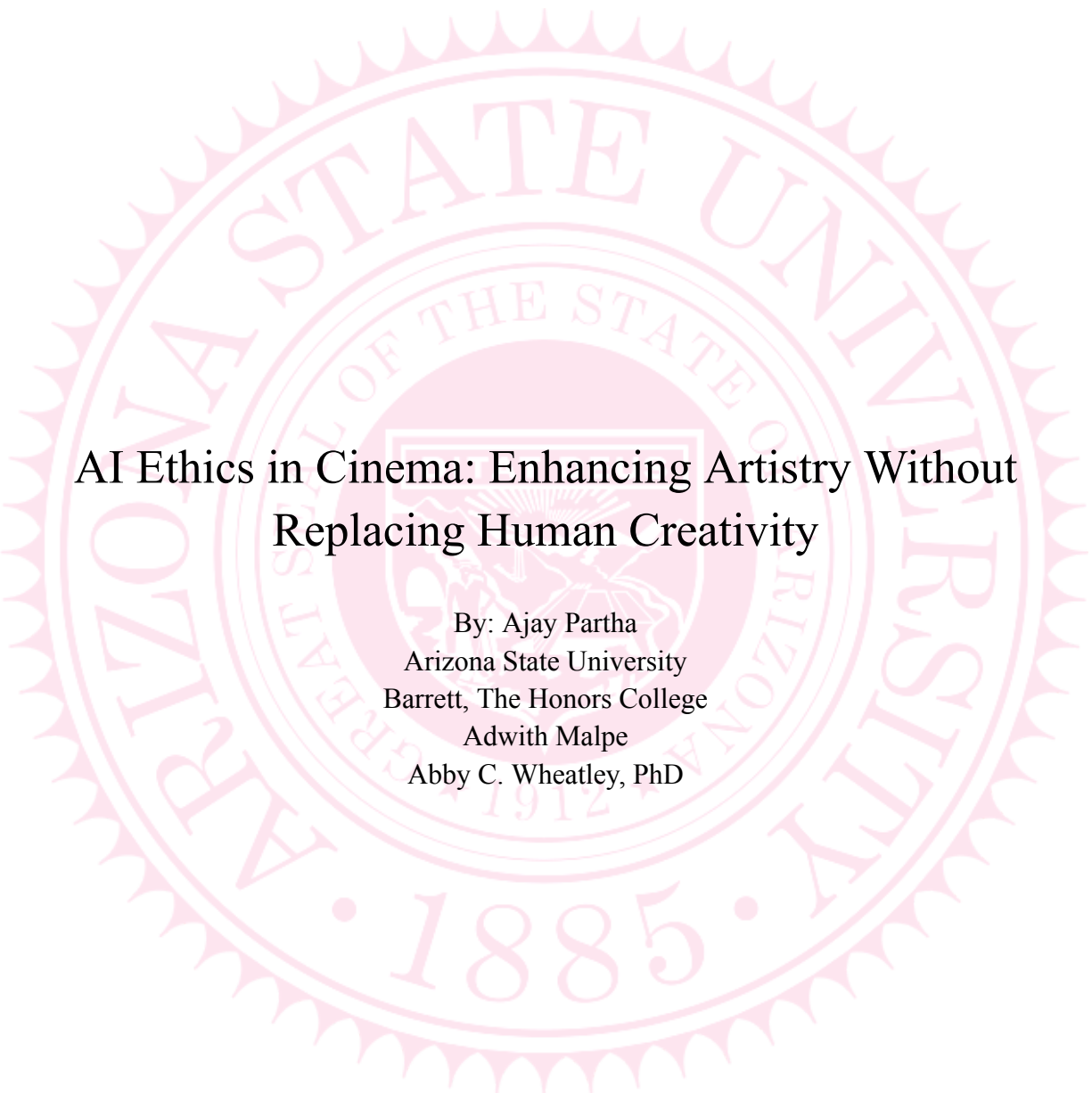


The background of the page features a large, faint, circular seal of Arizona State University. The seal contains the text "ARIZONA STATE UNIVERSITY" around the top, "1885" at the bottom, and "SCHOOL OF THE STATE" in the center. In the middle of the seal is a shield with a sunburst and a figure.

AI Ethics in Cinema: Enhancing Artistry Without Replacing Human Creativity

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Abstract

Artificial intelligence (AI) has rapidly progressed over the past couple of years, sparking worldwide concerns about whether these new technologies will take human jobs, especially in the film industry. This thesis delves deeper into the ethical implications of AI in filmmaking and its potential to change the film industry as we know it. AI technologies have found ways to streamline production processes, creating benefits such as cost efficiency for small-budget filmmakers. Although these developments can optimize pre-production, production, and post-production processes, it causes distress with factors such as ethical use of information, job security, and creative integrity. A qualitative methodology incorporating literature reviews and case studies of visual media, is used to explore the dual-sided sword AI can be: democratizing the filmmaking process while also making people question the authenticity of the new age of cinema. This study explores whether AI can maintain a human element in art without ruining authorship in AI-generated scripts, or the use of deep fakes. Recommendations are made to ensure transparency and consent are always prioritized when innovating, to maintain ethical integrity. This thesis argues that while AI can support filmmakers through creative inspiration, and streamlining certain processes, it should never completely replace human authorship. AI is there to complement, not supersede the authentic storytelling that defines film. By highlighting these obstacles, this thesis can add to the ongoing conversation about whether AI should be involved in cinema's future and our obligation to establish equitable and progressive regulations.

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Introduction

As a Computer Science major, there have been many moments where I have wondered whether jobs would still exist for me in 5 years with the emergence of AI. As these machine learning systems automate tasks that we thought only humans could achieve, I have questioned: Can these machine learning algorithms replace artistry or human storytelling? This question is now very apparent within the film industry, where AI has streamlined processes and created ethical concerns. From generating scripts to visual effects, AI has the chance to shake up cinema, but where will that place the people and fundamentals that have defined this form of art so far?

History of Technology in the Work Industry

Technological advancements have always created paradoxical situations. These discoveries bring new and efficient ways of expression while also making people question everything they knew up to that point. AI practices in film are not an exception to this theory and can be closely aligned with other technological disruptions in creative industries.

Going back to the 15th century, when we first saw the introduction of the printing press, this development changed storytelling forever by making this literature accessible to the public. This demonstrated that innovation can be influenced heavily by the simplicity of accessing existing knowledge. The printing press “clearly was one of the most significant access-cost-reducing inventions of the historical past.”¹ Of course, some people questioned the efficiency of the printing press (like scribes), as some saw it as a threat to their work, and they were the ones who saw the most loss. However, despite early fears, historical evidence has shown that there are long-term benefits. For instance, the data “shows that, on average, cities that

¹ Mokyr, Joel. “Long-Term Economic Growth and the History ...” *LONG-TERM ECONOMIC GROWTH AND THE HISTORY OF TECHNOLOGY*, 9 Aug. 2005, faculty.wcas.northwestern.edu/jmokyr/AGHION1017new.pdf.

adopted the press in the late 1400s grew 20 percentage points more and over 4 times faster than non-printing cities 1500-1600.”² This scenario demonstrates how the new technology initially scared the public, but new opportunities were introduced by adapting to and welcoming this change.

In the world of film, the introduction of computer-generated imagery (CGI) in the late 1900’s had a similar disruptive effect. CGI changed how films were animated, allowing for worlds that animators felt were impossible to achieve. Filmmakers who were not as experienced with technical tools could now incorporate more advanced animation techniques in their projects with the help of CGI.³ Although this technology brought a ton of world immersion, it created a lot of fear of job loss for hand-drawn animators. Those who could not transition to the new workflows struggled to find open positions. Nowadays, CGI has become standard in filmmaking and begs the question of whether the innovation was worth the disruption in the workforce.

AI has been the latest interference in innovation in the industry. This differs from CGI, as AI has been interfering with creativity in filmmaking. The new development has the ability to generate scripts, sequences, and even change the way actors talk. On top of that, it usually takes half the time. Filmmaking has never been this easy, but at what cost? How does automating film processes affect the lives of actors and artists? And can algorithms create art that humans resonate with?

Historically speaking, automation in creative industries has frequently led to a reshaping of work instead of creating complete disappearance. The introduction of photography in the 19th century sparked a lot of controversy. Many believed that a mechanically produced image could

² Dittmar, Jeremiah E. “Information Technology and Economic Change: The Impact of the Printing Press *.” *The Quarterly Journal of Economics*, vol. 126, no. 3, Aug. 2011, pp. 1133–1172, <https://doi.org/10.1093/qje/qjr035>.

³ Dous, Soumen. “THE EVOLUTION of VISUAL EFFECTS in CINEMA: A JOURNEY from PRACTICAL EFFECTS to CGI.” *Journal of Emerging Technologies and Innovative Research*, vol. 10, no. 11, Nov. 2023, www.researchgate.net/profile/Soumen-Das-22/publication/ Accessed 28 Jan. 2025.

not achieve the creative authenticity of paintings. Nowadays, photography is seen as an artistic medium with the capability to evoke human emotions. Another example of this is with CGI, which was once viewed as emotionally sterile. *Toy Story* was the first computer-animated feature film and demonstrated to the skeptical audience that digital animation could enhance artistic expression without eliminating human creativity.⁴ This movie was nominated for three Oscars and received an Academy Award for its exceptional contribution to motion pictures. History suggests that algorithms can face skepticism at first but as long as human artists are in the driving seats, their outputs can resonate with human audiences.

These historical shifts demonstrate the tension that comes with innovation. Although AI is a very powerful tool, it is optimal that its use lies in assisting filmmaking through inspiration, refinement, and facilitating smaller improvements. There should never be a point where it becomes the sole author of a script or completely replaces human-driven storytelling. Creativity in cinema thrives on human intuition and lived experiences which are aspects AI cannot authentically reproduce. Therefore, filmmakers must use AI to enhance instead of replace their vision, in order to preserve the uniqueness that comes from human craftsmanship.

Overview

This thesis focuses on the ethical challenges and opportunities initiated by the development of AI in filmmaking, looking closer at its impact on job security, creative integrity, audience engagement, and societal implications. Adopting AI technologies has allowed films to automate processes, create new visual effects, and generate new scripts. However, the

⁴ “Toy Story.” Oscars.org | Academy of Motion Picture Arts and Sciences, 21 Sept. 2015, www.oscars.org/collection-highlights/toy-story.

advancements do come with negative side effects regarding excessive reliance on AI for creative decisions with script writing and fewer job opportunities for artists.

The structure of this study addresses these questions through an ethical and archival lens. The introduction section provides some historical background for technological interferences in work industries, specifically the film industry, tracing back to the advent of the printing press and the addition of CGI. Historical context shows how new tech developments have always caused debates about whether they close or open new paths. Having this perspective creates a familiar understanding of the changes brought about by AI.

The next section delves into the ethical considerations around AI in cinema. Included in this section are the implications of AI-generated scripts, deepfake technology, and concerns about consent for storytelling. It specifically examines issues around intellectual property rights, questioning the ownership over AI-generated content and the human involvement needed to claim authorship. This section then explores ethical tensions around consent violations when actors' performances are curated without their permission. This discussion employs the help of recent legal cases to help safeguard individual identities within films. After exploring those concerns, they are analyzed to look closer at their effects on the job market and creative integrity for artists.

The following section, AI Methods in Filmmaking, investigates AI applications across the different stages of the production process. AI can be used to optimize scriptwriting and casting in the pre-production process. This can be through generating initial drafts, or suggesting variations to dialogue. The systems are able to analyze a vast amount of storytelling data to suggest changes to story arcs or how to improve the pacing. In addition, it can be used to de-age actors and synthesize actors' voices during the post-production process. All of these different

applications for AI raise the question of which ones should be allowed and which ones diminish creativity. The AI applications are looked at closer through three different films/series to identify what works versus what does not work.

The final few sections, Discussion and Recommendations, provide a broader overview of the findings and propose recommendations to ensure creative practices are conserved within the film industry. There are a few policy recommendations, such as transparency requirements for AI-generated content such as clear labeling of all content generated by AI. Additionally consent frameworks that govern using an actor's image and voice. The recommendations emphasize limiting AI's usage in creative processes to ensure artistic production is not fully automated. These safeguards ensure that artistic integrity, job security, and audience trust are all protected by maintaining a clear distinction between AI-assisted processes and human creativity.

Ultimately, this thesis contributes to the current debate on how AI can be responsibly integrated into the film industry. By examining the ethical risks and possible solutions, this thesis will help provide a comprehensive plan for balancing innovation with a sense of responsibility. This paper highlights both the positives and negatives of AI-assisted technologies in cinema and proposes how this powerful tool can be used ethically, as long as it stays in a supportive role. Human touch is critical when telling stories aimed at resonating with the audience, but that doesn't mean AI technologies can't contribute to this art form.

Methodology

This study uses a qualitative research methodology to investigate the ethical implications of AI in filmmaking. This approach was chosen because it can better understand subjective issues such as audience trust and creative integrity, which are not typically quantified.

The first key component of this research involved a thorough analysis of existing literature. Multiple academic articles, and legal documents, such as the 2023 Writers Guild of America's agreement on AI, were investigated to provide a deeper understanding of where the film industry currently stands with this innovative technology. The literature review is utilized for its ability to assess the ethical standing of AI across the different stages of filmmaking.

The next component of this research involved case studies. These different case studies were chosen to demonstrate real-world applications of AI in filmmaking. The three examples were evaluated by whether or not they preserved creative integrity and the opportunities they opened or closed for industry workers. They were also compared to one another, demonstrating the challenges and needs of filmmakers who are given different amounts of resources.

The research conducted utilized an ethical framework when investigating this problematic topic. This framework focused on areas such as ownership and consent, creative integrity, and job security, ensuring a more objective approach when looking at the consequences of AI in the making of films.

This thesis has a few limitations. Focusing on case studies could result in some overgeneralization of effects on the film industry as a whole. Also, since AI is constantly advancing, there might have been new developments since this research was conducted, which will require additional research.

Concerns in AI and Filmmaking

Ownership and Consent

The ownership of AI-generated content is one of the most prominent ethical dilemmas currently faced in the world of cinema. This includes the consent needed for using actors' image

and performances. AI systems require large datasets and information to have the capability to generate scripts and pictures, which makes people question intellectual property rights.

A question that people have been asking is if AI generates a script, who has the rights to the finished product? Lack of clarity around ownership rights for AI-generated content produces serious ethical and legal challenges, which therefore creates uncertainty for film creators. This deals with psychological and legal ownership. Psychological ownership deals with the feeling of owning something even if you don't legally own it. Legal ownership deals with your legal rights to use certain assets. A stronger sense of psychological ownership over a generative AI assistant has been linked with adding more detail to the user's prompts. For example, Joshi and Vogel “observed a strong positive correlation between Word Count and Psychological Ownership ($r_s = .72; p < .001$), suggesting that as participants wrote more, they felt more psychological ownership over their writing.”⁵ Users felt a stronger sense of authorship as they prompted the generative AI with more specifics and felt more comfortable using this in their professional work. However, higher psychological ownership does not necessarily equate to higher legal ownership. Even though users may feel connected to their AI creation, intellectual property laws still prevent AI content from qualifying for legal protection. Human involvement is required for authorship and copyright protection, which means AI-generated content can't be legally connected to an individual or organization. The question of how much human involvement is needed is still in the air. Stephen Thaler, a creator of an AI machine, “The Creativity Machine,” went to court to claim that any work that used this AI tool must designate the machine as the author. His appeal was rejected in court because the U.S. Copyright Office only registered an original work of authorship if human beings created the work.⁶ It showed that “not just AI which

⁵ Joshi, Nikhita, and Daniel Vogel. “Writing with AI Lowers Psychological Ownership, but Longer Prompts Can Help.” *ArXiv*, 22 Aug. 2024, arxiv.org/pdf/2404.03108.

⁶ Thaler, Stephen, et al. *UNITED STATES DISTRICT COURT for the DISTRICT of COLUMBIA*. 18 Aug. 2023.

can work independently from a human, even a machine that works with some or little human input cannot be said to be the author of the work produced by it.”⁷ These legal debates demonstrate the concern users face when working with AI and how institutions will govern their authorship rights. Courts and policymakers need to focus on this tension of ownership so that creative creators can have more transparency over their work when dealing with exploitation.

Another issue equally concerning is AI tools' ability to replicate actors without consent. AI-driven deepfake technology reveals serious ethical threats against actors by exploiting their identities, demonstrating the need for stronger consent protections and regulations. The word ‘deepfake’ alludes to “synthetic media generated using artificial intelligence techniques, particularly deep learning algorithms, to manipulate or fabricate audio, video, or images with startling realism.”⁸ The ability to create hyper-realistic versions of performers, typically without their approval, raises questions about what filmmakers could do with actors' likenesses. Although there aren't explicit laws against deepfakes, there are more broad legislation dealing with deceptive manipulated audio or visual images. For instance, California has enacted legislation allowing individuals to permanently block digital identity theft cases on any social media platform. In addition, Tennessee has added to the Personal Rights Protection Act with the “Ensuring Likeness, Voice and Image Security Act of 2024, to provide that every individual has a property right in the use of that individual's name, photograph, voice, or likeness in any medium in any manner.”⁹ These recent legal battles dealing with unauthorized AI uses of performers emphasize the urgent need for more explicit regulations to help safeguard individuals' images.

⁷ “View of AI-Generated Works: Copyright Ownership.” *Apricusjournals.com*, 2025, www.apricusjournals.com/index.php/jus-l-bulletin/article/view/171/96. Accessed 30 Jan. 2025.

⁸ Rath, Preeti, et al. *Unmasking Deepfakes: Understanding the Technology, Risks, and Countermeasures*. 21 Feb. 2024, <https://doi.org/10.1109/iciptm59628.2024.10563353>. Accessed 23 Sept. 2024.

⁹ “Deceptive Audio or Visual Media (“Deepfakes”) 2024 Legislation.” *Www.ncsl.org*, 10 Oct. 2024, www.ncsl.org/technology-and-communication/deceptive-audio-or-visual-media-deepfakes-2024-legislation.

This issue becomes highly concerning when AI-generated characters can mimic dead individuals, making it confusing whether it is homage or exploitation. The ethical ambiguity with AI grows stronger as film creators blur the line between a tribute and commercial manipulation, therefore questioning posthumous consent and dignity. This example can be seen in Peter Cushing's portrayal in *Rogue One: A Star Wars Story*. Although seeing his character "Tarkin" back on screen evoked nostalgia for many Star Wars fans, the act caused a lot of backlash. The main one was a lawsuit against Lucasfilm, the company that produced *Star Wars*, for using Cushing's likenesses without asking permission from his estate. Lucasfilm claimed that they had the right to his likeness through Cushing's *Star Wars* contract, but they ended up paying Cushing's estate approximately \$37,000.¹⁰ These practices not only pose the ethical question of consent but also question the authenticity of AI art if they can reduce actors to just commodities.

If audiences are not able to distinguish between AI-generated and real performances, this could lead to concerns about the authenticity of creative works, which will undermine the work made by filmmakers. The uncertainty produced by this process could diminish filmmakers and viewers' trust, leading to the elimination of emotional connection to films. To correctly address these issues, it is critical that there are clear standards of transparency around the usage of AI-generated content. This requires not only legal reforms but also discussions about the moral responsibilities of film production. Engaging industry leaders, filmmakers, and audiences in conversations about trustworthiness about the respectful use of rising technologies.

¹⁰ Voight, William. "Legal Battle over Star Wars Character Recreation." *The Legal Journal*, 19 Sept. 2024, thelegaljournal.com/2024/09/19/legal-battle-over-star-wars-character-recreation/.

Job Security

The labor market has seen a ton of reshaping due to the sudden rise of artificial intelligence. This has started concerns about job security and the future of the creative workforce. Unlike technological shifts in the past that have been strictly for assisting humans, AI's new ability to completely automate tasks threatens entire job markets specifically targeting entry-level jobs. Automation has the potential to change how the film process works and the types of roles that are available for human workers.

A significant change AI could cause in creative industries is the contraction of the job market. There will be a consolidation of creative work that will be turned into fewer and more centralized roles. Since AI technologies have the ability to streamline production processes, the demand for specialized labor will diminish, such as concept artists and screenwriters. An individual using AI tools can now achieve work that previously required an entire team. The result is a job market that favors a highly skilled workforce capable of overseeing AI systems, inadvertently eliminating entry-level and niche roles. A CEO of a top technology firm has said, "We will soon launch a robot that can perform tasks currently done by people with a high school education or less."¹¹ What will happen to workers without access to a sound education system?

Heavier reliance on AI will shift the skills needed to succeed in the film industry. This shift is extremely troubling as it marginalizes professionals who have the access to necessary training, silencing diverse voices and limiting the extent of storytelling. Reliance on AI could make performing creative and technical tasks not as important as knowing how to manage AI systems. There could be roles like "AI supervisors" that might take the sport of jobs such as assistant editors, or storyboard artists. This transition may create new and more innovative roles,

¹¹ West, Darrell M. *FUTURE of WORK : Robots, Ai, and Automation*. S.L., Brookings Inst, 2018.

but it also will cause obstacles for individuals who don't have the resources to adapt to the change. Typically, "countries with well established frameworks for skill training are positioned advantageously while others face livestock crises spiraling into unemployment, social unrest, and economic stagnation."¹² This new dynamic could cause a wider divide between individuals with access to technology and opportunities versus those struggling to find jobs in a shrinking market.

Economically speaking, reliance on AI can result in a higher concentration of wealth within a smaller number of film studios. This concentration is alarming because it undermines independent filmmakers and smaller studios, which restricts the range of storytelling and reduces potential opportunities for innovative narratives. When AI developments are only in automation, there is a massive tendency for wealth inequality to worsen. This trend is shown by, "the rise of automated manufacturing and logistics systems, where initial job losses among low-skilled workers lead to greater income disparities."¹³ This economic restructuring could affect the current wage structures and the broader economy that depends on filmmaking for jobs and imaginative statements.

Automation can reshape the job market. Before committing to these innovations, it is essential to see how they will affect the economy, wage distribution, and human expression. The adoption of AI could lead to a more systematic but less inclusive industry. Prioritization of efficiency at the expense of accessibility for creators with limited resources will make the film industry more exclusive and less inviting for talented minds. Ultimately, without planning and frameworks, automation within film can make this field incredibly resource-efficient but creatively impoverished.

¹² Challoumis, Constantinos. THE IMPERATIVE of SKILL DEVELOPMENT in an AI REVOLUTION. 13 Dec. 2024.

¹³ Liu, Fang, and Chen Liang. "Analyzing Wealth Distribution Effects of Artificial Intelligence: A Dynamic Stochastic General Equilibrium Approach." *Heliyon*, 1 Jan. 2025, pp. e41943–e41943, [www.cell.com/heliyon/fulltext/S2405-8440\(25\)00323-8](https://doi.org/10.1016/j.heliyon.2025.e41943), <https://doi.org/10.1016/j.heliyon.2025.e41943>. Accessed 2 Feb. 2025.

Creative Integrity

Can algorithms and data-driven processes that run AI tools create the emotional depth that has defined cinematic storytelling to this day? AI systems are trained on many datasets of existing cinematic works, but they rely on the same patterns to generate new content. Does this create homogenization of storytelling? When scripts are made solely through AI, a feedback loop occurs in which AI slightly changes popular tropes instead of introducing new ideas. Since AI ultimately lacks the ability to interpret subtle language clues such as sarcasm, the stories typically have little depth.¹⁴ These trends could lead to a decline in experimental films which have been some of the most loved in recent years. The commercially safe movies do have the most appeal to larger audiences. However, they can stifle the creative diversity in cinema.

Cultural bias and representation are another huge problem with using AI models. There seems to be a consistent pattern with large language models (LLMs) “the output of GPT models tends to resemble Western cultures when prompted without a specific cultural identity”¹⁵. Biases occur because AI models are trained on most Western-centric pieces of data. Effects of this training can be skewed representations of cultures and identities which promotes stereotyping rather than going against them. Human creators can consciously move towards a more inclusive narrative, whereas AI replicates patterns in their training data. This poses a considerable challenge for the recent progress in increasing representation on screen. Filmmaking needs diverse perspectives and is in danger of losing the collaborative nature of the industry. Unique insights should be prioritized over efficiency.

¹⁴ Rahman, Md Danish, and Mohammad Ashraf Ali. "AI in Video Production: From Script to Screen." *Media and AI: Navigating*: 49.

¹⁵ Tao, Yan, et al. "Cultural bias and cultural alignment of large language models." *PNAS nexus* 3.9 (2024): page346.

AI Methods in Filmmaking

Creative workflows across pre-production, production, and post-production have all been reshaped due to the integration of AI in this artistic process, changing the traditional methods that filmmakers are used to today. While AI has the ability to streamline a ton of the typical tasks seen in these processes, reducing production costs, these advancements raise concerns about human creativity, and the implications of these new technologies. To understand fully which uses of AI are ethically responsible, it is necessary to delve deeper into how AI-generated content is used at every step of the creative process.

AI in Pre-Production

In pre-production, AI-driven script generators can analyze a vast dataset filled with screenplays to create scripts within minutes. To put this in perspective, a “professional scriptwriter typically spends 12 weeks working on a script after receiving a commission from a studio or a producer.”¹⁶ These AI models are typically based on transformer architectures and generative AI, such as GPT-4, that allows users to receive written responses from their prompting. Generative AI utilizes large language models (LLMs), which learn language rules and patterns to respond to tasks such as writing content and summarizing text.¹⁷ The data that is used to train the LLM “has been scraped from the internet, though it is unclear exactly which sources have been included in the corpus for each AI tool,”¹⁸ thus creating concern with accuracy

¹⁶ Sun, Peiming. “A Study of Artificial Intelligence in the Production of Film.” *SHS Web of Conferences*, vol. 183, 2024, p. 03004, www.shs-conferences.org/articles/shsconf/abs/2024/03/shsconf_icpahd2024_03004/shsconf_icpahd2024_03004.html, <https://doi.org/10.1051/shsconf/202418303004>.

¹⁷ “What Is a Large Language Model (LLM)? - University of Arizona Libraries.” Arizona.edu, 2023, ask.library.arizona.edu/faq/407985.

¹⁸ “Generative AI in Academic Writing.” The Writing Center • University of North Carolina at Chapel Hill, July 2023, writingcenter.unc.edu/tips-and-tools/generative-ai-in-academic-writing/.

and bias. Since the infrastructure for the LLM - internet devices - is available constantly, this is what makes generation so fast. As stated before, due to generative AI being trained on datasets of current projects, it poses the question of whether they can truly capture the emotional depth of characters in human-written scripts and create new stories in the process.

Visual conceptualization has also seen AI advancements through new technologies that generate storyboards from descriptions. Creating generative models like DALL-E and Midjourney has allowed us to take descriptive prompts and turn them into visual representations for the storyboarding process. DALL-E can turn text into images utilizing transformer models, neural networks that break down texts by their patterns and paint their pixels based on context. But, this translation from text to image is inherently complex due to the flexibility of these interpretations. As some researchers have shown, “the task of translating text to images is underspecified: a single caption generally corresponds to an infinitude of plausible images, so the image is not uniquely determined.”¹⁹ This can be useful in many situations; however, it can create some confusion in the storyboarding process. Since storyboarding requires “cohesion between images, the perception of the narrative logic of a film, especially a popular one with iconic characters, can be altered through DALL-E’s potential for aesthetic mimicry.”²⁰ For example, in a research experiment DALL-E is used to create a new storyboard using images from *Twilight*, but decides to make the film more vibrant and positive (*O’Meara and Murphy, 2023*). Although generative models can create visually appealing content with a much smaller budget, their inability to maintain a consistent narrative that is authentic raises worries about reliability. The unpredictability that comes with this technologically advanced method, may create complications with collaboration, and disrupt the artistic vision by confusing the artists.

¹⁹ Liu, Anting, and Shichen Zhang. "Taking Text to Image for Spin via DALL-E." 2023,

²⁰ O’Meara, Jennifer, and Cáit Murphy. "Aberrant AI creations: co-creating surrealist body horror using the DALL-E Mini text-to-image generator." *Convergence* 29.4 (2023): 1070-1096.

This demonstrates that generative models are able to produce high-quality images without the need for large budgets, but are they able to create authentic storyboards like humans?

AI in Production

During the production phase, AI applications are used for motion capture and on-set animation, which can change how live performances are interpreted in digital media. Motion capture (MoCap) systems have now utilized artificial intelligence to convert human movement into data that is easily consumed by digital cameras. MoCap is utilized in cinema by filmmakers so they can create realistic digital characters and movements for more immersive storytelling. There are two types of MoCap systems: sensor-based and optical. Sensor-based systems include inertial measurement units to track motion in real-time, whereas optical systems utilize reflective markers and high-speed cameras to capture performance information. Both of these cases demonstrate AI algorithms' potential to smooth out captured data for the creation of digital characters. As AI technology advances significant shifts have occurred reducing the amount of hardware to complete the same tasks. Researchers have found, “traditional motion capture systems rely on a large number of sensors and specialized equipment, whereas AI technology has the potential to capture motions by analyzing standard video inputs.”²¹ Still, high-end optical systems are only used by major studios due to the high costs of advanced cameras, but sensor-based solutions are much more affordable for independent filmmakers.

On-set automation technologies have also emerged recently, helping with camera quality, lighting control, and production scheduling. AI-enhanced cameras can change focus and exposure dynamically, while smart lighting systems can change the intensity of the color in the

²¹ Xi, Chen, and Jeanhun Chung. "A Case Study on AI-Driven< DEEPMOTION> Motion Capture Technology." *International Journal of Internet, Broadcasting and Communication* 16.2 (2024): 87-92.

scene. The downside of this advancement is that it displaces old-school technical roles in film and eliminates a human aspect in cinematography.

AI in Post-Production

Post-production contains AI's most visibly alternating applications through computer-generated imagery (CGI), the de-aging process, and voice synthesis. CGI, which is AI-driven, utilizes deep learning techniques, like generative adversarial networks, to create and refine visual effects (VFX). Specific techniques, such as super-resolution and image inpainting, help strengthen footage for VFX. On the other hand, specialized software like NVIDIA's Omniverse is critical in simulating realistic environments. A type of AI-driven CGI technology that has become popular recently is de-aging technologies. These technologies were deployed in Martin Scorsese's 2019 film, *The Irishman*, to recreate younger versions of Robert De Niro, Joe Pesci, and Al Pacino. De-aging actors is not an easy process, and they typically demand high-performance workstations with powerful GPUs and substantial memory capacities. Although a lot of performance power is needed, this process is advantageous. *The Irishman's* deaging made the 80-year-old actors look 40, although the complaints were that the actors didn't move like 40-year-olds. This process eliminated the need for heavy makeup and prosthetics, saving the actors plenty of time during filming.²²

Advancements in vocal tools, such as AI-based voice synthesis systems, help immerse the audience in a film. Companies such as Respeecher and DeScript have been able to create technologies that can mimic natural intonation and pacing through deep-learning models. These engines delve deep into datasets of voice recordings to produce dialogue and sound effects nearly

²² Essien, Essien, Awni Kalkat, and Sonika Lamichanne. "Paradigm Shift of Visual Effects (Vfx) in Hollywood Apocalypses: Analyses of Advanced Cgi Artistry in Selected Films." Available at SSRN 4925458.

identical to the real thing. This easily accessible voice alternating development makes advanced techniques accessible to filmmakers with various budgets. Still, they raise ethical concerns about whether they are being used to recreate actors' voices without consent.

Case Studies

Sigma_001:

Sigma_001 is a 2023 short film that exemplifies how independent filmmakers can leverage AI technology to innovate and overcome budgetary constraints. This film follows an engineer at a tech company who entails the work of a journalist to report on what he believes is the first sentient AI. Throughout the film's production journey, the creators deployed AI tools to help with streamlining creative workflows. This helped raise efficiency but posed a question to the industry: Should they be more lenient to independent filmmakers when using AI?

In pre-production, the writing team utilized generative AI to help guide the development of the narrative. They used early versions of ChatGPT to ensure that "Sigma" (the sentient AI) acted in ways that felt true to real-world technology. In addition, they used tools like DALL-E and Runway to generate concept art and turn it into storyboards. This multifaceted use of AI allowed the filmmakers to explore different creative directions with nowhere near the costs it would typically take.

AI continued to the production phase by incorporating affordable sensor-based motion capture technology. The crew discovered that "drawing concept art, tightening the script, adjusting the camera angles, etc.—could be done more effectively and

economically with the help of AI.”²³ AI chatbots were also used by actors so that they could better understand their characters and build on their backstories that were not apparent in the script. The use of these technologies helped the production team enhance the precision of the camera work without needing the funding of a large filmmaking team. Integration of AI also demonstrated that these developments not only save time and money but can also reinforce storytelling decisions.

Post-production for *Sigma_001* pronounced the effectiveness of AI in creating advanced visual effects (VFX). The team used tools like DeepFaceLabs to deepfake characters and quickly shift them from different personas. Halleck, the director and creator of *Sigma_001*, has been reported saying, “there’s a real way to integrate [AI] into what we would already be doing, which is mimicking the process of big studio movies as close as I can with the money I have.” (*Sebastopol Times* 2024) Demonstrating that these AI tools could be game-changing for small-budget films. Another tool incorporated into this short was AI voice generation from ElevenLabs to ensure an authentic artificial voice for “Sigma.” Effects that would have cost this team tons of time and money have been made much simpler through AI.

The result of this AI usage was an enchanting short, but still, it received backlash for utilizing these tools. When Halleck was invited to the Sebastopol Center for the Arts (SebArts) for a Q&A about this film, he was grilled about his usage of AI. Many were worried that AI would make this art form dull and unimaginative. In response, Halleck focused more on his experience with AI in filmmaking and supported improving safeguards of technology in film. When asked about the over-reliance on AI in the future

²³ Sebastopoltimes.com, 2025, www.sebastopoltimes.com/p/ai-film-screened-to-impassioned-audience. Accessed 5 Feb. 2025.

of cinema, Halleck responded, “I can see a future where it can democratize the process and allow people to have more access to some of the things that were once more difficult to achieve, which hopefully will result in better and cooler, more diverse art, because you can get more cool, diverse voices.” (*Sebastopol Times* 2024) Halleck believes that AI is beneficial when it comes to being a tool, not over depending on it. He also states, “I mention the names of my crew not only to give them credit, but to illustrate my idea that it takes real artist to bring the film to its final vision.”²⁴

Halleck’s film highlights a world where technology and human creativity can exist coincidentally, ensuring that AI tools remain supportive but do not replace the human touch needed for this form of art. Now the film’s usage of generative AI, such as ChatGPT, helped refine the dialogue and character behavior for the sentient AI but this should not be the standard for all films. The reason why this worked was due to the fact that this film was about artificial intelligence. However, there were AI tools that were used in the creation of this film that are ethically responsible that should be utilized in independent filmmaking. During production of this film, AI-supported motion capture allowed these filmmakers to achieve precise camera work, which usually takes a good amount of manpower and budget. Democratizing access to advanced film techniques is one of the strongest benefits of AI-supported tools, allowing filmmakers to overcome budget limitations and fostering an industry with richer artistic voices.

Secret Invasion:

Marvel Studios recently joined the group of filmmakers that have used AI in their creative process with their 2023 series, *Secret Invasion*. During the premiere of *Secret*

²⁴ “Sigma_001 | Quinn Halleck.” *Quinn Halleck*, 2023, www.quinnhalleck.com/sigma. Accessed 5 Feb. 2025.

Invasion, the opening sequence involving a digitally altered portrayal of Samuel L. Jackson caught viewers' attention the most. After the premiere, Ali Salim, director and executive producer, revealed to the audience that the visuals in the opening sequence were generated with the help of artificial intelligence. This decision put Marvel at the forefront of the debate on AI in cinema. Marvel's hesitation in being transparent with openly disclosing AI usage emphasizes why clear communication is needed in the film industry to address ethical concerns.

The Marvel production team utilized generative AI and AI diffuser modes to refine the opening credit scene, which they felt played along with the themes of the series. These AI tools were not used to replace human artists in the creative world but instead were utilized to allow them to achieve more. Method Studios, who were the creators of the opening scene, clarified, "AI is just one tool among the array of tool sets our artists used. No artists' jobs were replaced by incorporating these new tools; instead, they complemented and assisted our creative teams."²⁵ This new approach allowed Method Studios to utilize AI for efficiency while sustaining artistic standards. The AI-driven tools allowed for quick alternatives for the scene, allowing the team to make faster decisions on the visual effects.

The reveal of AI involvement in this Marvel series caused serious debate. Many believed this would turn into an over-reliance on technology in creative work, causing impersonal and subordinate films to be produced. Many studies show that if audiences know AI creates a piece of work, they automatically think it is inferior regardless of the

²⁵ Giardina, Carolyn. "“Secret Invasion” Opening Using AI Cost “No Artists’ Jobs,” Says Studio That Made It (Exclusive)." The Hollywood Reporter, 22 June 2023, www.hollywoodreporter.com/tv/tv-news/secret-invasion-ai-opening-1235521299/.

final product.²⁶ Research regarding AI paintings versus human-created paintings showed, “if participants believed the painting involved more effort in its creation than other paintings, participants liked paintings made by humans more than those made by AI.”²⁷ These perspectives on AI make studios wary of revealing their usage, just as Marvel released their reliance on AI after the premiere.

Marvel responded to this backlash by defending the transparency of AI. The creative team emphasized how keeping AI in a supportive role is vital to avoid job security concerns. Marvel preaches transparency; however, not everyone on the visual development team for *Secret Invasion* knew about incorporating AI into their work. Jeff Simpson, a concept artist, tweeted, “I’m devastated, I believe AI to be unethical, dangerous and designed solely to eliminate artists careers,”²⁸ when he learned how AI was used in *Secret Invasion*.

Marvel's *Secret Invasion* shows how AI can be used to increase the scope of a product and how important it is to openly state the AI-driven tools used in the process. These technologically advanced tools can expand the efficiency of production processes but transparent communication is vital. Addressing AI usage allows creators to navigate ethical concerns and inform audiences on what to expect. The transparency also helps with maintaining innovation with responsibility, ensuring that AI is supporting human creativity instead of replacing it.

²⁶ Noti-Victor, Jacob. "Regulating Hidden AI Authorship." (2024).

²⁷ Bellaiche, Lucas, et al. “Humans versus AI: Whether and Why We Prefer Human-Created Compared to AI-Created Artwork.” *Cognitive Research: Principles and Implications*, vol. 8, no. 1, 4 July 2023, [cognitiveresearchjournal.springeropen.com/articles/10.1186/s41235-023-00499-6](https://doi.org/10.1186/s41235-023-00499-6), <https://doi.org/10.1186/s41235-023-00499-6>.

²⁸ Horton, Adrian. “Marvel Faces Backlash over AI-Generated Opening Credits.” *The Guardian*, 21 June 2023, www.theguardian.com/tv-and-radio/2023/jun/21/marvel-ai-generated-credits-backlash.

The Brutalist:

The Brutalist (2024) is an intense and visually striking film that explores the realities of being an outsider during post-war America. The film has been praised and has garnered early buzz for the awards season, making it one of the strongest contenders for multiple Oscars. *The Brutalist* used artificial intelligence in multiple ways during the production process to achieve a level of realism that would have been costly otherwise. This AI usage has made many critics believe that this film is not deserving of any Oscars, however minimal AI usage can help enhance an actor's authentic performance without getting rid of the human artistry.

Dávid Jancsó, an editor on this film, revealed to tech magazine Red Shark News that AI tools from Respeecher were deployed on Adrien Brody and Felicity Jones's Hungarian dialogue.²⁹ He also disclosed that generative AI was used in the final sequence to create architectural drawings and finished buildings in a style that resembled Brody's character, László Tóth. The sound team initially tried Automated Dialogue Replacement (ADR), a process of recording dialogue after filming concludes to substitute the initial lines on Brody's and Jones' dialogue, but they concluded that it still didn't sound the way they wanted it to. AI software was then used with Jancsó's voice to fix the problematic dialect. This incorporation of AI tools helped the production team believe that they created a finished product that was as historically accurate as they could make it.

Director Brady Corbet has been very vocal about defending the use of AI in his movie. Corbet has gone on record saying, “Adrien and Felicity’s performances are completely their own... Respeecher technology was used in Hungarian language dialogue

²⁹ Pennington, Adrian. “Why Epic Period Drama Movie the Brutalist Was Shot on VistaVision.” *Redsharknews.com*, 2023, www.redsharknews.com/why-epic-period-drama-movie-the-brutalist-was-shot-on-vistavision.

editing only, specifically to refine certain vowels and letters for accuracy.”³⁰ His sentiments are supported by lead actor Adrien Brody, who believes that Corbet is a conscious filmmaker and “there was no technology implemented that takes work from people.”³¹ The director and cast realize the threat AI can pose to the film industry but ultimately support specific AI methods that can help deepen the storytelling process.

The film's approach to producing *The Brutalist* has sparked debate among critics about whether it deserves to win awards. Many believe that the reliance on AI goes directly against the craftsmanship celebrated by the Oscars. The Academy currently has an optional disclosure for AI use, but now they are moving towards “making disclosure mandatory in the 2026 Oscars rules, which are expected to be published in April.”³² The case of *The Brutalist* demonstrates the importance of transparency and careful integration of AI to maintain an artist’s authenticity. When used minimally, AI tools can help make an actor’s performance more captivating and allow the film to be more genuine.

Discussion and Analysis

Findings from both ethical considerations and technical methods demonstrate how integrating AI in filmmaking presents opportunities and challenges for the creative process. The biggest revelation here is the need to preserve human artistry by ensuring AI remains as a tool that is strictly supportive instead of dominant.

³⁰ Roxborough, Scott. ““The Brutalist” Director Responds to AI Backlash.” *The Hollywood Reporter*, 20 Jan. 2025, www.hollywoodreporter.com/movies/movie-news/the-brutalist-ai-backlash-adrien-brody-1236113015/.

³¹ Canfield, David. “Adrien Brody Doesn’t Want AI Controversy to Detract from “the Brutalist”: “We All Know the Truth.”” *Vanity Fair*, 4 Feb. 2025, www.vanityfair.com/hollywood/story/adrien-brody-brutalist-interview-awards-insider?utm_source=twitter&utm_medium=social&utm_campaign=dhtwitter&utm_content=null. Accessed 8 Feb. 2025.

³² Giardina, Carolyn. “Oscars Consider Requiring Films to Disclose AI Use.” *Variety*, 7 Feb. 2025, variety.com/2025/artisans/news/oscar-consider-requiring-films-disclose-ai-use-brutalist-1236299063/.

AI-driven tools, when used judiciously, have the potential to expand the creative worlds seen in cinema today. They are the most efficient when they are used as an assistant instead of their own entity. For instance, when generating storyboards for the filmmaking process, generative AI could help brainstorm ideas before a concept artist ultimately adds to the one they believe is best. These processes can help refine artistic visions without eliminating human decision-making. Generative AI utilizes sources from the internet to curate their images, so how is this different from the artist looking at other artists' work for inspiration? However, as critics argue, over-reliance on AI could lead to a diminished learning environment. If filmmaking procedures become fully automated, where will artists be able to exemplify their creativity? Humans should always be in the driving seat making the final call, and AI should only play a role in certain parts of film creation.

AI assistance has proven invaluable in refining motion capture data and generating creative ideas in the pre-production phase. Still, its use in generating full animations, deepfakes, and complete scripts raises serious concerns. These applications strip away the human touch needed for storytelling in film. Deepfakes blur the line between reality and fabrication, which raises serious ethical challenges. Fully AI-written scripts have not been able to truly capture complex character dynamics and put writers' jobs at risk.

In contrast, utilizing AI as a supportive tool has shown that technology augments, especially with smaller filmmakers. Independent producers can now access cutting-edge AI-driven tools that reduce production costs, helping to level the playing field. Certain tools, like sensor-based motion capture technology, have expanded independent creators' creative capabilities and fueled innovation. Technology democratization has helped ensure more diverse voices in the film industry.

To fully take advantage of the benefits of AI, there must be a secure regulatory framework articulating transparency when using AI. A clear outline of practices and ethical guidelines is key to mitigating adverse reactions surrounding AI usage. This will allow audiences to realize the supportive effectiveness these advancements can bring rather than shortcuts that eliminate jobs. In the realm of acting, there also needs to be clear guidelines on the amount of AI allowed for an actor's performance. Allowing too many AI-generated enhancements to the portrayal of characters dilutes the human talent that comes with performing. There needs to be a boundary that prevents AI from entirely replacing actors' performances to ensure that awards like the Oscars continue to honor human creativity.

Recommendations

Transparency in AI Usage

Establishing clear guidelines that encourage transparency in AI usage is key in ensuring that it serves as a tool instead of a surrogate for human creativity. Firstly, both studios and filmmakers must acquire a mandatory disclosure policy that will clearly communicate AI's involvement to the audience. These disclosures are key in promoting Responsible AI (RAI) principles, which involve accountability and transparency. RAI helps ensure that the development and usage of AI systems are ethical and safe for everyone. To further transparency, these disclosures should be easy to understand and user-friendly. In addition to adding these disclosures in legal documents, filmmakers should include dedicated statements in the film's closing credits and marketing materials. Employing the help of UX/UI designers could help design adequate disclosures that would be presented to the audience at the correct time. This could entail standardized labels for AI-generated content, like dietary labels on food items or

debriefs on the film's webpage. In an MIT panel, AI experts weighed in on this debate, and eighty-four percent of these panelists supported the idea that companies disclose AI usage.³³ They believe the disclosure retains consumer trust and builds confidence in the progression of AI. Transparency in where AI is used on an actor's performance should also be revealed. In order to maintain authenticity in actors' performances we need to create a limit on how much their acting can be altered by these advanced tools. For research papers, "a general rule of thumb is that no more than fifteen percent of your paper should be quoted material."³⁴ This allows for researchers to convey their own ideas without just reiterating facts already stated by people before them. This general rule should be carried over to the filmmaking process, but perhaps a bit stricter around 10%. Enforcing this would allow for acting performances to stay original while allowing for technologically advanced tools to enhance artistry. Creating this assurance through clear disclosure guidelines allows us to progress as a creative society without losing sight of what makes films meaningful.

Consent and Rights Management

The contracts for AI-driven filmmaking have to emphasize a complete framework that clearly explains the scope of artificial intelligence usage, ensuring that everyone working on the project has consented to AI modifications to their work. These contributors include animators, actors, and directors, and each one deserves to know the changes being made to their ideas. The agreements will detail precisely what aspects will be altered by AI. For instance, specifying certain visual enhancements or post-production effects. These accords will also state what parts

³³ Renieris, Elizabeth M., David Kiron, and Steven Mills. "Artificial Intelligence Disclosures are Key to Customer Trust." *MIT Sloan Management Review (Online)*, 2024, pp. 1-4. *ProQuest*, <https://login.ezproxy1.lib.asu.edu/login?url=https://www.proquest.com/scholarly-journals/artificial-intelligence-disclosures-are-key/docview/3118850624/se-2>.

³⁴ APA Direct Quotations. Monmouth University, 28 May 2013, www.monmouth.edu/resources-for-writers/documents/apa-direct-quotations.pdf/.

of the film are untouchable, granting workers security with their work on the project. Informed consent is so necessary nowadays to regain the trust of creative artists since AI has been associated with getting rid of their jobs and work. Another concept that should be addressed in these contracts is the usage duration of consent. This would outline any of the limitations on the future usage of AI-generated work. Actors would be able to delineate how long their likeness could be used for that franchise. Finally, compensation mechanisms must be added to contracts to ensure that AI-generated work utilizing actors' likenesses is not left unnoticed. Thankfully, the 2023 Sag-AFTRA strike has ensured that “time spent creating a digital replica counts as work time and is paid accordingly.”³⁵ However, it should be more precise on how and where these replicas can be used beyond initial production. This includes marketing, merchandising, and future work. Contracts that are more specific about how AI is utilized will allow the process to be more transparent and welcoming for workers.

Job Security Protections

The outcome of the 2023 SAG-AFTRA strike secured protections for writers against AI. For example, the contract states, “Writers have agency: They can choose to use generative AI in their writing process, but cannot be forced to do so by the studios.”³⁶ These more vigorous measures should be extended to all film creators, including animators, cinematographers, and production designers. Cinematographers and production designers must be granted control over how these AI tools will be utilized in visual effects and should be given strict oversight over these AI-generated elements. All of these roles are vital in making films, and they should all be

³⁵ *Regulating Artificial Intelligence*. SAG-AFTRA, 2023, www.sagaftra.org/sites/default/files/sa_documents/AI%20TVTH.pdf.

³⁶ Kinder, Molly. “Hollywood Writers Went on Strike to Protect Their Livelihoods from Generative AI. Their Remarkable Victory Matters for All Workers.” *Brookings*, 12 Apr. 2024, www.brookings.edu/articles/hollywood-writers-went-on-strike-to-protect-their-livelihoods-from-generative-ai-their-remarkable-victory-matters-for-all-workers/.

protected from the chance of their jobs being taken away by AI. Contracts should guarantee that AI use in all departments is disclosed, allowing for a transparent work community.

Support for Independent Filmmakers

Independent filmmakers are so crucial for continued innovation in the film industry. They produce new perspectives and don't focus on the money-making factor of cinema. However, it is becoming harder and harder for these filmmakers to make movies as they face significant resource constraints compared to big studios. Promoting access to ethical AI tools is vital to even out the playing field and bring more diverse voices to the cinema. Indie filmmakers utilizing these AI-driven tools have shown the industry the discrepancy between funding for movies. Some key strategies for independent filmmakers' voices to be heard more are increasing funding initiatives or grant programs, fostering collaborative networks on AI use, and establishing clear standards for AI in film production. The major corporations that are developing these AI tools should be the ones who are sponsoring grant programs that are targeted at independent filmmakers. Companies like OpenAI, ElevenLabs, and Runway should allocate funds that are obtained from people using their tools to reduce-cost licenses for their AI software to independent projects. These organizations could also partner with film institutions to host training sessions, educating filmmakers on the ethical use of AI. Having AI companies actively pursuing the ethical use of their tools will create a more positive attitude towards this innovation. Leveling the playing field with ethical AI tools can increase innovation in cinema and encourage a culture of experimentation.

Conclusion

Artificial intelligence has had a transformative impact on filmmaking and will continue to have one. This impact has the potential to create tons of new opportunities but also poses ethical dilemmas on how to approach them. The integration of AI in the production process has been proven to democratize the creative process, allowing independent filmmakers to produce work that could be compared to major studios. However, these technological advancements also raise concerns about job security for creative artists and the ethical management of digital likenesses.

Evidence, as seen in this thesis, proves that AI should be embraced as a tool rather than a complete replacement for the human touch. By regulating AI, the emotional depth of this art form will stay at the forefront of storytelling and won't be abolished. It is essential that people who are wary of AI in film stay advocating for transparent practices and regulatory frameworks that can safeguard the rights of creative workers. Yet, these same outspoken critics should open their minds to the possibility of AI equally playing the playing field in cinema and promoting innovation. A balanced approach where human artistry and technology can work hand in hand produces a film industry that is more inclusive and ethically responsible.

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